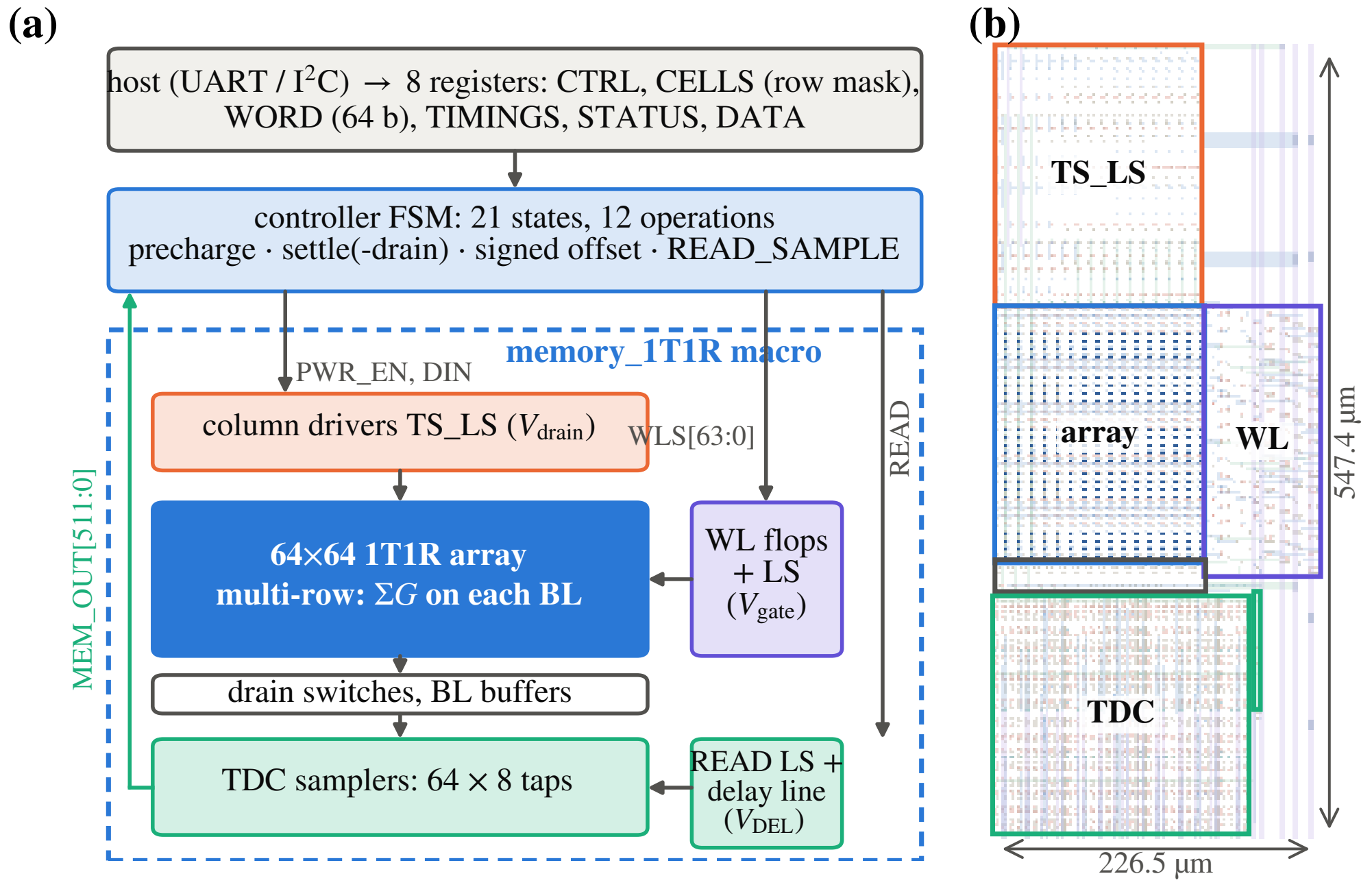
(a)
host (UART / I²C) → 8 registers: CTRL, CELLS (row mask), WORD (64 b), TIMINGS, STATUS, DATA
controller FSM: 21 states, 12 operations
precharge · settle(-drain) · signed offset · READ_SAMPLE
memory_1T1R macro
PWR_EN, DIN
column drivers TS_LS (V_{drain})
WLS[63:0]
READ
MEM_OUT[511:0]
64×64 1T1R array
multi-row: ΣG on each BL
WL flops + LS (V_{gate})
drain switches, BL buffers
TDC samplers: 64 × 8 taps
READ LS + delay line (V_{DEL})
(b)
TS_LS
array
WL
TDC
547.4 µm
226.5 µm